

# PH8001: COMPUTATIONAL BIOLOGY, EXPERIMENTAL DESIGN AND DATA SCIENCE

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## Effective Term

Summer Term 2025

## Part I Course Overview

### Course Title

Computational Biology, Experimental Design and Data Science

### Subject Code

PH - Infectious Diseases and Public Health

### Course Number

8001

### Academic Unit

Infectious Diseases and Public Health (PH)

### College/School

Jockey Club College of Veterinary Medicine and Life Sciences (VM)

### Course Duration

One Semester

### Credit Units

3

### Level

R8 - Research Degree

### Medium of Instruction

English

### Medium of Assessment

English

### Prerequisites

Nil

### Precursors

Nil

### Equivalent Courses

Nil

### Exclusive Courses

Nil

## Part II Course Details

### Abstract

This course aims to make the postgraduate students

- a) equipped with the fundamental knowledge in computational biology and data science;
- b) prepared with practical skills and appropriate logic to analyze molecular or numerical data, including data processing, visualizing, interpreting, and hypothesizing; and
- c) capable of designing biomedical/veterinary projects/experiments rationally. Python will be the main programming language used in the practical sessions.

#### Course Intended Learning Outcomes (CILOs)

| CILOs |                                                                                                                          | Weighting (if app.) | DEC-A1 | DEC-A2 | DEC-A3 |
|-------|--------------------------------------------------------------------------------------------------------------------------|---------------------|--------|--------|--------|
| 1     | Explain and apply the concepts, logic and algorithms underlying the commonly used bioinformatics tools                   | 10                  | x      | x      |        |
| 2     | Attain the ability of performing data mining for the -omics data using proper tools/parameters under a Linux environment | 40                  | x      | x      | x      |
| 3     | Design a biomedical/biological experiment regarding statistical and biological factors                                   | 15                  | x      | x      | x      |
| 4     | Perform explanatory data analysis with Python                                                                            | 15                  | x      | x      | x      |
| 5     | Apply supervised machine learning models to biological data for regression or classification                             | 20                  | x      | x      | x      |

#### A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

#### A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

#### A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

#### Learning and Teaching Activities (LTAs)

| LTAs |          | Brief Description                                                                                                                                     | CILO No.      | Hours/week (if applicable) |
|------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------|
| 1    | Lectures | Students will engage in interactive lectures to understand the fundamental philosophy/ algorithms and apply appropriate tools for - omics data mining | 1, 2, 3, 4, 5 |                            |

|   |                                   |                                                                                                                                                                                                                                                                                                           |               |                  |
|---|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------|
| 2 | Hands-on practical tasks          | Students will perform problem-based practices to<br>1) strengthen the understanding of the principles, algorithms, or philosophy underlying the models/tools; and<br>2) apply bioinformatics or data mining tools/ models on biomedical/ veterinary problems and interpret the results                    | 1, 2, 3, 4, 5 |                  |
| 3 | Take-home assignments and reports | Students will complete project-based assignments to consolidate the understanding of bioinformatics or data mining tools/models and proficiency in performing the analyses                                                                                                                                | 2, 3, 4, 5    | Out of classroom |
| 4 | Q&A sessions                      | Students will participate in Q&A sessions to<br>1) clarify concepts or correct the misunderstanding of the principles, algorithms or philosophy underlying the models/tools;<br>2) expand the scope of knowledge in biomedical data mining;<br>3) apply methods/tools learned to the real-world problems. | 1, 2, 3, 4, 5 |                  |

**Assessment Tasks / Activities (ATs)**

|   | ATs                  | CILO No. | Weighting (%) | Remarks ("- " for nil entry)                                                                                                                       | Allow Use of GenAI? |
|---|----------------------|----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1 | Classroom assessment | 2, 3     | 30            | Formative assessment will be carried out to evaluate students' comprehension and improve the learning outcomes (aligns with ILOs 1, 2, 3 4 and 5). | Yes                 |

|   |             |               |    |                                                                                                                                                                                                                          |     |
|---|-------------|---------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2 | Assignments | 1, 2, 3, 4, 5 | 70 | These tasks are designed to evaluate the ability of assessing pros and cons of different tools/ models and the ability of applying them to realistic biomedical or veterinary problems (aligns with ILOs 2, 3, 4 and 5). | Yes |
|---|-------------|---------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|

**Continuous Assessment (%)**

100

**Assessment Rubrics (AR)****Assessment Task**

Classroom assessment (30%) (for students admitted before Semester A 2022/23 and in Semester A 2024/25 &amp; thereafter)

**Criterion**

Students will apply their competence in the content of both the theoretical and practical components.

**Excellent**

(A+, A, A-) High

**Good**

(B+, B, B-) Significant

**Fair**

(C+, C, C-) Moderate

**Marginal**

(D) Basic

**Failure**

(F) Not reaching basic levels

**Assessment Task**

Assignments (70%) (for students admitted before Semester A 2022/23 and in Semester A 2024/25 &amp; thereafter)

**Criterion**

Students will appraise their competence in the key concepts and algorithms of commonly used bioinformatics tools, construct biological/veterinary experiments based on the principles taught, and utilize the techniques/tools learned to solve specific biological problems.

**Excellent**

(A+, A, A-) High

**Good**

(B+, B, B-) Significant

**Fair**

(C+, C, C-) Moderate

**Marginal**

(D) Basic

**Failure**

(F) Not reaching basic levels

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**Assessment Task**

Classroom assessment (for students admitted from Semester A 2022/23 to Summer Term 2024)

**Criterion**

The comprehension of the contents in both the theoretical and practical parts

**Excellent**

(A+, A, A-) High

**Good**

(B+, B) Significant

**Marginal**

(B-, C+, C) Basic

**Failure**

(F) Not even reaching marginal levels

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**Assessment Task**

Assignments (for students admitted from Semester A 2022/23 to Summer Term 2024)

**Criterion**

- 1) The comprehension of the key concepts and algorithms in the commonly used bioinformatics tools; ability to design a biological/ veterinary experiment based on the principles taught in this course;
- 2) the ability to solve some specific biological problems using the techniques/ tools learned/ recommended in this course.

**Excellent**

(A+, A, A-) High

**Good**

(B+, B) Significant

**Marginal**

(B-, C+, C) Basic

**Failure**

(F) Not even reaching marginal levels

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## Part III Other Information

Keyword Syllabus

Bioinformatics, computational biology, sequence analysis, -omics data mining, bioinformatics software, Linux operation, Python programming, experimental design, data visualization, multivariate analysis, supervised machine learning, unsupervised machine learning, exploratory data analysis, predictive modeling, causal inference

## Reading List

### Compulsory Readings

| Title |                                                                                                                                                                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Biological sequence analysis, R. Durbin, S.Eddy, A. Krogh, G.Mitchison, <a href="https://pdfs.semanticscholar.org/2ed5/d6b35f8971fb9d7434a2683922c3bfcc058e.pdf">https://pdfs.semanticscholar.org/2ed5/d6b35f8971fb9d7434a2683922c3bfcc058e.pdf</a> |
| 2     | Python Data Science Handbook, Jake VanderPlas, <a href="https://github.com/jakevdp/PythonDataScienceHandbook">https://github.com/jakevdp/PythonDataScienceHandbook</a>                                                                              |

### Additional Readings

| Title |                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------|
| 1     | Deep Learning, Yoshua Bengio, MIT Press, ISBN10 0262035618                                                                       |
| 2     | The Book of Why: The New Science of Cause and Effect, Judea Pearl and Dana Mackenzie, ISBN-10: 046509760X                        |
| 3     | How to Create a Mind: The Secret of Human Thought Revealed, Ray Kurzweil, Penguin Books; 7/28/13 edition, ISBN-10: 9780143124047 |